OPTIMUM COACHING & GUIDE

কম্পিউটার সাইন্স এন্ড ইঞ্জিনিয়ারিং

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Syllabus of Admission Test

Computer Fundamentals: Basic Organization and Functional Units; Hardware, Software, Software and its applications.

Basic Programming: Basic program structure and IDE, pseudo code, header files, data types operators, variables and expressions; Input and Output; control statement and decision making, loop sructure, arrays, strings, pointers, function, recursion, pointers and dynamic memory allocation, arrays, strings, multidimensional array; structures, unions enumerations, file I/O system, linking, library functions.

Object Oriented Programming (C++/Python): Object oriented programming basics, class & object, Properties of object oriented programming.

Discrete Mathematics: Probability theory, Counting, tree & graph.

Data Structure: Array, pointer, linked list, stack, queue, searching.

Database Management: Database system concept, data models, query languages.

Software Development: System analysis, system design, system implementation.

Operating system: Scheduling, memory management, I/O system, File system.

Microprocessor & Microcomputer: 8086 Architecture, bus systems, instruction sets, assembly language, interrupt control, interfacing chips, memories, secondary storage devices.

Data Communication & Computer Network: Transmission media, modulation/demodulation, Multiplexing, OSI and TCP/IP model, network topologies, network protocols, intemets, IP addressing.

Computer Graphics: Graphics pipeline, modeling, animation, rendering, relation to computer vision and image processing, geometric transformations, Rasterization.

Digital Electronics: Number systems, Boolean algebra, Logic gates, Combinational and sequential logic circuits, Flip-Flops, Registers, Counters, VLSI, AID and D/A Converters.

Electronic Devices and Circuits: Semiconductor Diodes and Rectifiers, BJT, JFET, MOSFET, CMOS, Amplifiers, Oscillator circuits, Photo Electric devices, Multivibrator, Time Base Circuits.

Basic Electricity: Circuit laws, DC circuits, AC circuits.